

Lecture 2:

Machine learning basics and supervised learning

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GRA4160: Predictive Modelling with Machine Learning

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Plan for today:

1. Machine learning basics
2. Linear regression
3. Supervised learning with k -nearest neighbors
4. Exercise: Spam filtering with naive Bayes

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Exploring the Fundamentals of Machine Learning

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- ▶ **Reinforcement learning**: Training agents to make decisions towards a goal. Example: Robot navigation.

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- ▶ It involves input-output pair training, like predicting house prices from their size and location.
- ▶ Key algorithms include:
 - ▶ **Linear regression:** For continuous output prediction.
 - ▶ **k -nearest neighbors:** Classifies based on similarity.
 - ▶ **Decision trees:** Maps decisions and their possible consequences.
 - ▶ **Neural networks:** For complex pattern recognition.

- ▶ A core concept in ML: **Bias** is how much your predictions systematically deviate from the true values.
- ▶ **Variance** is how sensitive your model is to fluctuations in the training set.
- ▶ **High bias** models:
 - ▶ Underfit the data (oversimplified).
 - ▶ Have consistent but inaccurate predictions.
- ▶ **High variance** models:
 - ▶ Overfit the data (too complex).
 - ▶ Perform well on training data but poorly on new data.
- ▶ Goal: **Find a balance** that minimizes overall error.

Three supervised learning algorithms

We will look closer at three supervised learning algorithms:

- ▶ Linear regression
- ▶ k -nearest neighbors
- ▶ Naive Bayes (exercise)

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- ▶ Be mindful that the linear form might not always capture real-world complexity, so model diagnostics are crucial.

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- ▶ Typical practice: Choose k via **cross-validation**.

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- ▶ **Non-parametric models** (e.g., kNN):
 - ▶ Make fewer assumptions about data distribution.
 - ▶ Number of parameters can grow with the dataset.
 - ▶ Flexible but often more computationally expensive and can overfit without careful tuning.

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- ▶ Extremely **fast** and easy to implement, making it a good first model in many classification tasks.
- ▶ We will explore its spam-filtering application, highlighting the simplicity and speed of this approach.